

GONGFAN CHEN

Assistant Professor
Dept. of Engineering Technology and Construction Management
University of North Carolina at Charlotte
237 Smith Hall, 319 Library Ln, Charlotte, NC 28223
Email: gchen12@charlotte.edu
Website: <https://hgv523.github.io/chatlab>
LinkedIn: <https://www.linkedin.com/in/gongfan-chen/>

RESEARCH INTEREST

Human-technology interactions; Large language model (LLM) applications for construction knowledge communications; Blockchain and smart contract applications to enhance project collaboration reliability; Decentralized computing framework for task-level automation; Workflow simulation to enhance project situation awareness; Computer vision algorithms for automated site monitoring and inspections; and Pedagogical framework for AI implementation in STEM education.

EDUCATION

- **North Carolina State University**, Raleigh, NC, U.S.
Ph.D. in Civil Engineering Aug 2019 – Aug 2023
M.S. in Electrical and Computer Engineering Aug 2021 – May 2023
- **University of Michigan**, Ann Arbor, MI, U.S.
M.S. in Civil Engineering Aug 2017 – May 2019
- **North China Electric Power University**, Beijing, China
B.S. in Engineering and Management Sept 2013 – June 2017

PROFESSIONAL APPOINTMENTS

- **Assistant Professor**, University of North Carolina at Charlotte, August 2025 – present
- **Visiting Research Scholar**, North Carolina State University, August 2025 – present
- **Postdoctoral Research Scholar**, North Carolina State University, August 2023 – August 2025
- **Graduate Research Assistant**, North Carolina State University, August 2019 – August 2023
- **Graduate Teaching Assistant**, North Carolina State University, August 2019 – August 2023
- **Graduate Teaching Assistant**, University of Michigan, August 2018 – December 2018
- **BIM Engineer**, China Railway 24th Bureau Group Co., Ltd., May 2018 – August 2018
- **BIM Lecture**, North China Electric Power University (Beijing), January 2016 – June 2016
- **Research Assistant**, North China Electric Power University (Beijing), September 2014 – June 2017

PUBLICATIONS

Referred Journal Articles:

- [J10]. Chen, G., Jaselskis, E., Tamer, A., and Folz, A., (2025). "Using a Large Language Model Powered Framework for Automatic Risk Identification and Mitigation from Past Claims and Supplemental Agreements." *Journal of Computing in Civil Engineering* (Accepted).
- [J9]. Diaz, B., Chen, G., Delgado, C., and Jaselskis, E. (2025). "[Supporting Generative AI Literacy: Exploring the](#)

[Pedagogical Roles Students Assign ChatGPT and Impact on Course Grades.](#)” *Comunicar*, 33(82), 46–61.

- [J8]. Shafaay, M., Alqahtani, F. K., Alsharef, A., and Chen, G. (2025). [Modeling construction cost overrun risks at the FEED stage for mining projects using PLS-SEM.](#) *Journal of Asian Architecture and Building Engineering*, 1–17.
- [J7]. Chen, G., Alsharef, A., Ovid, A., Albert, A. and Jaselskis, E. (2025). [“Meet2Mitigate: An LLM-powered framework for real-time issue identification and mitigation from construction meeting discourse.”](#) *Advanced Engineering Informatics*, 64: 103068.
- [J6]. Chen, G., Alsharef, A., and Jaselskis, E. (2024). [“Construction Jobsite Image Classification Using an Edge Computing Framework.”](#) *Sensors*, 24 (20):6603.
- [J5]. Chen, G., M. Liu, H. Li, S. M. Hsiang, and A. Jarvamard. (2023). [“Motivating Reliable Collaboration for Modular Construction – A Shapley Value-Based Smart Contract.”](#) *Journal of Management in Engineering*, 39 (6): 04023042. (Editor’s Choice)
- [J4]. Chen, G., M. Liu, Y. Zhang, Z. Wang, S. M. Hsiang, and C. He. (2023). [“Using Images to Detect, Plan, Analyze, and Coordinate a Smart Contract in Construction.”](#) *Journal of Management in Engineering*, 39 (2): 1–18.
- [J3]. He, C., M. Liu, Y. Zhang, Z. Wang, S. M. Hsiang, G. Chen, W. Li, and G. Dai. (2023). [“Space-Time-Workforce Visualization and Conditional Capacity Synthesis in Uncertainty.”](#) *Journal of Management in Engineering*, 39 (2): 04022071.
- [J2]. Chen, G., H. Li, M. Liu, S. M. Hsiang, and A. Javanmardi. (2022). [“Knowing what is going on – a smart contract for modular construction.”](#) *Canadian Journal of Civil Engineering*, 50 (3): 210-223. (Invited paper).
- [J1]. He, C., M. Liu, Y. Zhang, Z. Wang, S. M. Hsiang, G. Chen, and J. Chen. (2022). [“Exploit Social Distancing in Construction Scheduling: Visualize and Optimize Space–Time–Workforce Tradeoff.”](#) *Journal of Management in Engineering*, 38 (4): 1–15.

Referred Conference Articles:

- [C8]. Chen, G., Pedraza, M., Albaloul, R., Albert, A., and Jaselskis, E. (2025). “Developing a Fit-for-Purpose Best Practice Knowledge Handbook using Generative AI.” *CSCE Construction Specialty Conference / ASCE Construction Research Congress (CRC) 2025*.
- [C7]. Chen, G., and Liu, Y. (2025). “Leveraging Large Language Models for Voice-Activated Robotic Teleoperation in Construction.” *ASCE International Conference on Computing in Civil Engineering 2025*.
- [C6]. Mapare, V., Liu, Y., Zheng, D., and Chen, G. (2025). “Enhancing the Worker-Robot Interaction for Construction Task Operation: A Preliminary Study Using Large Language Model-based Methods.” *ASCE International Conference on Computing in Civil Engineering 2025*.
- [C5]. Chen, G., Alsharef, A., and Jaselskis, E. (2024). “Enabling Real-Time AI Applications in Construction Site Safety: A Novel Edge Computing Approach for Nails Detection Under Resource-Constrained Environment.” *ASCE International Conference on Computing in Civil Engineering 2024*.
- [C4]. Chen, G., Alsharef, A., and Jaselskis, E. (2024). “Integrating Sensor-Empowered Federated Learning and Smart Contracts for Automatic Sustainable Infrastructure Management.” *ASCE International Conference on Computing in Civil Engineering 2024*.
- [C3]. Chen, G., C. He, S. Hsiang, M. Liu, and H. Li. (2023). [“A mechanism for smart contracts to mediate production bottlenecks under constraints.”](#) *31st Annual Conference of the International Group for Lean Construction (IGLC)*. Lille, France.
- [C2]. Li, H., G. Chen, M. Liu, S. M. Hsiang, and A. Jarvamardi. (2023). [“Situation awareness based smart contract for modular construction.”](#) *Proceedings of the Canadian Society of Civil Engineering Annual Conference 2021*, 363–373. Singapore: Springer Nature Singapore. (Best Student Paper for Awardee: Gongfan Chen).
- [C1]. He, C., M. Liu, Z. Wang, G. Chen, Y. Zhang, and S. M. Hsiang. (2022). [“Facilitating Smart Contract in Project Scheduling under Uncertainty—A Choquet Integral Approach.”](#) *Construction Research Congress 2022*, 930–939.

TEACHING AND MENTORING EXPERIENCE

Instructor

- **CMET-2105 Plan Reading**, University of North Carolina at Charlotte (Fall 2025)

Teaching Assistant:

- **Undergraduate-Level Course**
 - **CE365 – Construction Equipment and Method**, NC State University (Spring 2021)
 - **CE463 - Construction Estimating, Planning, and Control**, NC State University (Fall 2019, Fall 2020, Fall 2021)
 - **CE469 – Construction Engineering Project**, NC State University (Spring 2021, Spring 2022, Spring 2023)
 - **CEE431 – Construction Contracting**, University of Michigan (Fall 2018)
- **Graduate-Level Course**
 - **CE564 – Legal Aspects of Contracting**, NC State University (Spring 2022, Spring 2023, Spring 2024)
 - **CE592 – CII Best Practices**, NC State University (Fall 2021)
 - **CE592 – Global Construction Practices**, NC State University (Fall 2023)
 - o Class Coordinator with five different universities from four countries, including Tsinghua University (China), Tianjin University (China), Chulalongkorn University (Thailand), University of Lagos (Nigeria), and NC State University (US).

Mentees

- **Graduate Student**
 - Michael Pedraza (Civil Engineering), graduate student, NC State University
 - Roya Albaloul (Civil Engineering), graduate student, NC State University
 - Myat Mon Aye (Civil Engineering), graduate student, NC State University
- **Undergraduate Student**
 - Yongxuan Zhang (Mathematics), NC State University
 - Shilun Zhao (Mechanical Engineering), NC State University
 - Hongxu Liu (Engineering Management), North China Electric Power University
- **Associated Schools of Construction (ASC) Region 2 Faculty Mentor**
 - David Daildly (senior, Construction), NC State University
 - Sidney Perkinson (senior, Construction), NC State University
 - Ruby Cain (junior, Construction), NC State University
 - Austin Booth (junior, Construction), NC State University

INDUSTRY EXPERIENCE

- **BIM Engineer**, China Railway 24th Bureau Group Shanghai Railway Co., Ltd., May 2018 – August 2018, Beijing
 - Led a technical team to construct underground utility 3D BIM models for Beijing DaXing airport municipal roads.
 - Established a 3D roaming for BIM models and deployed in VR environments to tackle potential clash detection.

AWARDS

- **Travel Grants Award**, NSF Workshop at I3CE2024 Conference, 2024
- **ASCE Journal of Management Editor's Choice**, North Carolina State University, Raleigh, 2023

- **People's Choice Award**, NCSU CCEE 3-Minute Thesis Competition, 2022
- **Best Student Paper Award**, Canadian Society of Civil Engineering Annual Conference, 2021
- **Graduate Merit Award (GMA) Fellowship**, North Carolina State University, Raleigh, 2019
- **Second Place Award**, "Rush Hour" Business Decision-Making and Management Competition, Role: team captain, Beijing, China, 2017
- **Exchange Scholarships**, National Chi Nan University, Puli, Taiwan, 2015
- **First Place Award**, National Business Road Management Decision-making Competition, Role: team Captain, Shanghai, China, 2015
- **Department Outstanding Student**, North China Electric Power University (NCEPU), Beijing, China, 2014 – 2017

CERTIFICATE & MEMBERSHIP

- **Construction Industry Institute (CII) Professional Development Certificate**, Nashville, TN, 2025
- **PL-300 Microsoft Power BI Data Analyst Certification Training Program**, Raleigh, NC, 2025
- **Generative AI for Business with Microsoft Azure OpenAI Program**, Raleigh, NC, 2024
- **IBM Data Science Professional Certificate**, 2022
- **BIM Engineering Modeling Technical Certificate**, Certificate No. 160090200652, Beijing, China, 2016
- **BIM Engineer (Modeler) Professional License**, Certificate No. BJLJMBIMGCS00690, Beijing, China, 2016
- **Associate Member**, American Society of Civil Engineering (ASCE)

PRESENTATION

Invited Lectures:

- **Blockchain-Enabled Smart Contract for the Construction Industry**, CE 4130 Construction Contracts, California State Polytechnic University-Pomona, 2025
- **Leveraging BERTopic Modeling to Identify Risk Categories for Claims and Supplementary Agreements**, CE514- Decision Making and Risk Management in Construction, King Saud University, Saudi Arabia, 2024

Poster Presentation:

- **Meet2Mitigate: An LLM-Powered Framework for Real-Time Issue Identification and Mitigation from Construction Meeting Discourse**, Applied AI in Engineering & Computer Science Symposium, North Carolina State University, 2024

Conference Presentation:

- **Enabling Real-Time AI Applications in Construction Site Safety: A Novel Edge Computing Approach for Nails Detection Under Resource-Constrained Environment**, ASCE International Conference on Computing in Civil Engineering, Pittsburgh, Pennsylvania, 2024
- **Integrating Sensor-Empowered Federated Learning and Smart Contracts for Automatic Sustainable Infrastructure Management**, ASCE International Conference on Computing in Civil Engineering, Pittsburgh, Pennsylvania, 2024
- **A Fit-for-Purpose (F4P) Handbook for Effective Implementation of CII Best Practices**, Construction Research Congress 2024, Des Moines, Iowa, 2024

- **Insights for Effective Risk Management in Transportation Projects**, Transportation Research Board (TRB) Annual Meeting, Washington DC, 2024
- **Situation Awareness Based Smart Contract for Modular Construction**, Canadian Society of Civil Engineering Annual Conference, Niagara Falls, Ontario (Online), 2021

University Competition:

- **Situation Awareness Smart Contracts for Construction Management**, CCEE 5th Annual Three Minute Thesis Competition, North Carolina State University, 2022

PROFESSIONAL SERVICE

- **Reviewer**, International Journal of Construction Management, 2025 – Present
- **Reviewer**, Automation in Construction, 2025 – Present
- **Reviewer**, ASCE Journal of Computing in Civil Engineering, 2024 – Present
- **Reviewer**, Results in Engineering, 2024 – Present
- **Reviewer**, ASCE Journal of Management in Engineering, 2023 – Present
- **Reviewer**, Construction Innovation, 2022 - Present
- **Reviewer**, Sustainability, 2024 – Present
- **Reviewer**, Buildings, 2024 – Present
- **Reviewer**, Big Data and Cognitive Computing, 2024 – Present
- **Reviewer**, Applied Science, 2024 – Present
- **Reviewer**, Intelligent Infrastructure and Construction, 2024 – Present
- **Reviewer**, Peer J Computer Science, 2024 – Present
- **Reviewer**, Discover Artificial Intelligence, 2024 – Present
- **Reviewer**, Peer-to-Peer Networking and Application, 2024 – Present
- **Reviewer**, ASCE International Conference on Computing in Civil Engineering, Pittsburgh, Pennsylvania, 2025
- **Reviewer**, ASCE International Conference on Computing in Civil Engineering, Pittsburgh, Pennsylvania, 2024
- **Reviewer**, International Symposium on Automation and Robotics in Construction (ISARC), Dubai, UAE, 2021
- **Associate Member**, American Society of Civil Engineering (ASCE)

SKILLS

- **Language Proficiency:** Professional English, Native Chinese
- **Programming Languages:** Python, JavaScript, SQL, Matlab, R, SAS, Julia, Solidity
- **Tools/Software:** Power BI, Revit, Navisworks, BIM 360, HoloLens, Unity 3D, AutoCAD, Synchro, Lumion, GIS, Bentley, Primavera, Bluebeam Revu, Ganache, MetaMask, TensorFlow, PyTorch, Arena, Stroboscope
- **Hardware:** Raspberry Pi, Jetson Nano, wearable sensors